

ORIGINAL ARTICLE

Tumor mutational burden and immune infiltration as independent predictors of response to neoadjuvant immune checkpoint inhibition in early TNBC in GeparNuevo

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Background: The predictive value of tumor mutational burden (TMB), alone or in combination with an immune gene expression profile (GEP), for response to neoadjuvant therapy in early triple negative breast cancer (TNBC) is currently not known, either for immune checkpoint blockade (ICB) or conventional chemotherapy.

Patients and methods: We obtained both whole exome sequencing and RNA-Seq data from pretreatment samples of 149 TNBC of the recent neoadjuvant ICB trial, GeparNuevo. In a predefined analysis, we assessed the predictive value of TMB and a previously developed immune GEP for pathological complete remission (pCR).

Results: Median TMB was 1.52 mut/Mb (range 0.02–7.65) and was significantly higher in patients with pCR (median 1.87 versus 1.39; $P = 0.005$). In multivariate analysis, odds ratios for pCR per mut/Mb were 2.06 [95% confidence intervals (CI) 1.33–3.20, $P = 0.001$] among all patients, 1.77 (95% CI 1.00–3.13, $P = 0.049$) in the durvalumab treatment arm, and 2.82 (95% CI 1.21–6.54, $P = 0.016$) in the placebo treatment arm, respectively. We also found that both continuous TMB and immune GEP (or tumor infiltrating lymphocytes) independently predicted pCR. When we stratified patients in groups based on the upper tertile of TMB and median GEP, we observed a pCR rate of 82% (95% CI 60% to 95%) in the group with both high TMB and GEP in contrast to only 28% (95% CI 16% to 43%) in the group with both low TMB and GEP.

Conclusions: TMB and immune GEP add independent value for pCR prediction. Our results recommend further analysis of TMB in combination with immune parameters to individually tailor therapies in breast cancer.

Key words: neoadjuvant immune checkpoint inhibition, triple negative breast cancer, tumor mutational burden, exome sequencing

INTRODUCTION

Combining immune checkpoint blockade (ICB) with chemotherapy increased response rates in patients with metastatic triple negative breast cancer (mTNBC).^{1,2} For early triple negative breast cancer (TNBC), recent results of the neoadjuvant randomized phase II GeparNuevo study suggest that the addition of durvalumab to anthracycline/taxane-based neoadjuvant chemotherapy increases the rate of pathological complete remission (pCR), mainly in the

subgroup of patients treated with durvalumab alone before the start of chemotherapy.³

In several cancer types, predictive factors for ICB response include programmed death-ligand 1 (PD-L1) expression,¹ gene expression profiles (GEP) of infiltrating immune cells,⁴ as well as microsatellite instability (MSI)⁵ and tumor mutational burden (TMB).^{4,6–9} Based on results from bucket trials of ICB in metastatic cancer, Cristescu et al. recently proposed a combination of TMB with immune GEP as a predictor of response.¹⁰

In early TNBC, immune GEP and tumor infiltrating lymphocytes (TIL) have both prognostic and predictive values for response to chemotherapy.^{11,12} A high TMB could increase neoantigens inducing immune response. Indeed, in pooled analyses of diverse cancer types, overall a positive

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association of TMB and tumor immune cell infiltration has been observed.¹³ Similarly, comparison of breast cancer subtypes reveals that both high TMB and high immune infiltration are more frequent in TNBC compared with luminal breast cancers.^{14,15} However, within the TNBC subgroup, no such positive correlation of TMB and immune infiltration was detected, which may be attributed to immunoediting.¹⁶

The predictive value of TMB, alone or in combination with immune GEP, for response to neoadjuvant therapy in early TNBC is currently not known, either for ICB or conventional chemotherapy. It is especially not clear whether TMB will add to the predictive value of immune GEP. Therefore, our goal was to study these two parameters in pretreatment samples of TNBC from the neoadjuvant ICB trial GeparNuevo. We found that both parameters add independent value for pCR prediction for chemotherapy with and without ICB. Our results recommend further analysis of TMB in combination with immune parameters to individually tailor therapies in breast cancer.

METHODS

All analyses were carried out according to the reporting recommendations for tumor marker prognostic studies (REMARK) criteria.¹⁷ A CONSORT type diagram¹⁸ of the flow of samples through the study is shown in [supplementary Figure S1](#) (available at *Annals of Oncology* online).

Patients

The details of the GeparNuevo study (NCT02685059) have been described in a recent publication.³ In total, 174 patients were treated with nab-paclitaxel followed by dose-dense epirubicin/cyclophosphamide chemotherapy and were randomly assigned to either simultaneous treatment with durvalumab or placebo. The study was approved by the ethics committee and the competent authority. All patients provided written informed consent for study conduct, biomaterial collection, and analysis. In the window phase, durvalumab/placebo alone was given 2 weeks before the start of nab-paclitaxel; 117 of the 174 participated in this window phase. Randomization was stratified by stromal tumor infiltrating lymphocyte (sTILs). Patients with primary cT1b-cT4a-d disease, centrally assessed sTILs, and confirmed TNBC were included. The primary objective was pCR (ypT0 ypN0). The pCR rate with durvalumab was 53.4% (42.5% to 64.1%) versus placebo 44.2% (33.5% to 55.3%; $P = 0.287$), corresponding to odds ratio (OR) = 1.45 (0.80–2.63, unadjusted Wald $P = 0.224$). The durvalumab effect was significant in the window cohort (pCR 61.0% versus 41.4%, OR = 2.22, 1.06–4.64, $P = 0.035$; interaction $P = 0.048$).³ The complete set of molecular parameters was successfully obtained only for a subset of the patients: TMB was available for 149 patients ([supplementary Figure S1](#), available at *Annals of Oncology* online).

TIL scoring and PD-L1 immunohistochemistry

TIL scoring and PD-L1 immunohistochemistry were carried out as previously described.³ In brief, PD-L1 status was

determined using the Ventana SP263 antibody. We evaluated the PD-L1 expression as the percentage of tumor cells with membranous staining and percentage of TILs with membranous or cytoplasmic staining (relative to total TILs).³ Stromal TILs were evaluated based on the standardized guidelines of the international TIL working group.¹⁹ When TILs are included as continuous scores in logistic regression analysis, odds ratios are presented per 10% increase.

Whole exome sequencing for TMB and mutational signatures

Whole exome sequencing (WES) was conducted on fresh-frozen pre-therapeutic core biopsies and patient-matched blood samples with Illumina HiSeq 4000 (Illumina Inc., San Diego, CA). High-quality data were obtained for 149 (85.6%) of the 174 patients from the trial ([supplementary Figure S1](#), available at *Annals of Oncology* online). Full details on the methods are given in the [supplementary Methods](#) (available at *Annals of Oncology* online). For TMB calculation, we used the final list of 12 314 non-synonymous single-nucleotide variants (SNVs) and indels, obtained after filtering, together with an effective DNA coverage of 46 Mb to determine mutations per Mb for each sample. Mutational signatures were identified as described by Alexandrov et al.²⁰ R package SomaticSignatures was employed to estimate the proportion of each sample's mutations that have been assigned to each of the 21 mutational signatures.²¹

RNA sequencing

RNA sequencing was carried out on formalin-fixed paraffin-embedded (FFPE) tissue using an HTG EdgeSeq instrument (HTG Molecular Inc., Tucson, AZ) with the HTG EdgeSeq Oncology Biomarker Panel (2549 genes) based on an RNA-extraction-free chemistry and a nuclease protection assay.²² The tumor area was marked on a slide stained with hematoxylin and eosin and the area of invasive breast cancer recorded. From a corresponding unstained slide, 15 mm² tissue was scraped and used for library preparation according to the manufacturer's instructions. Libraries were quantified, pooled, and sequenced on an Ion Torrent S5 instrument (Thermo Fisher Scientific, Waltham, MA). Count tables were generated using the HTG parsing tool. Full details are given in the [supplementary Methods](#) (available at *Annals of Oncology* online). RNA sequencing data from pre-therapeutic cores were available for 159 of the 174 patients ([supplementary Figure S1](#), available at *Annals of Oncology* online).

Molecular subtyping from RNA-Seq and immune GEP

Molecular subtyping from RNA-Seq was carried out using the AIMS method²³ (details in [supplementary Methods](#), available at *Annals of Oncology* online). We also evaluated a predefined immune GEP predictive for neoadjuvant response that was created from a list of genes we previously identified in the GeparSixto study (GeparSixto immune signature: *CXCL9*, *CCL5*, *CD8A*, *CD80*, *CXCL13*, *IDO1*, *PDCD1*, *CD274*, *CTLA4*, *FOXP3*).⁴ The genes *CD21* and *IGKC*

were omitted because they were not covered by the mRNA sequencing assay. The immune GEP was calculated as the mean of the expression of the genes from the signature.

Aggregation of WES and HTG-RNA-Seq data

The analyses of the pseudonymized genomic datasets were carried out fully blinded to any clinical or pathological sample information. The final blinded whole exome sequencing (WES) and HTG-RNA-Seq datasets were transferred to German Breast Group (GBG) headquarters. WES and RNA-Seq were available for 149 and 159 patients, respectively, with both data available for 136 patients ([supplementary Figure S1](#), available at *Annals of Oncology* online). A comparison of the complete trial cohort, the WES cohort, and the WES+RNA-Seq cohort is provided in [supplementary Table S1](#) (available at *Annals of Oncology* online).

Statistical analysis

All clinical data, including age, stage, histological grade, treatment arm, window treatment, TILs, PD-L1, and pCR (ypT0 ypN0), were extracted from the clinical study database at GBG headquarters and represent central assessment. Both TMB and the immune GEP (GeparSixto immune signature) from RNA-Seq were used as continuous parameters in univariate and multivariate logistic regression. In addition, we present results of predefined dichotomized TMB and GEP for illustration of pCR frequencies and comparison with previous studies. For GEP, a median split of the cohort was used. TMB was dichotomized at the upper tertile (≥ 95 mutations) based on the previously suggested pan-cancer cutoff for ICB response prediction (>100 mutations).¹⁰ Pearson's chi-square and Fisher's exact test were applied to assess associations between categorical parameters. To analyze the predictive value of molecular markers for pCR, univariate and multivariate logistic regression models adjusted for prespecified variables were used. The R software environment (version 3.3.2, <http://www.r-project.org/>) and SPSS²⁴ (<http://www.ibm.com/>) were used for all analyses. All confidence intervals (CI) reported are 95%. All *P*-values are two-sided and $P \leq 0.05$ was considered as significant.

RESULTS

TMB and correlation with pCR in the GeparNuevo trial

We successfully obtained high-quality WES data from the tissue of 149 (86%) of the 174 patients from the GeparNuevo study ([supplementary Table S1](#), available at *Annals of Oncology* online, compares clinical data of the WES cohort and the complete study cohort). Median TMB was 1.52 mut/Mb (range 0.02–7.65). As shown in [Table 1](#), we found significantly higher median TMB values in older patients ($P < 0.001$) and a trend for higher values in patients with higher stage of cancer ($P = 0.086$). Numerically, we saw lower values for median TMB in tumors with high numbers of TILs and in those without PD-L1 expression, but this was not significant. Patients with a pCR displayed a significantly

Table 1. Differences in tumor mutational burden (TMB) according to clinical parameters of the GeparNuevo samples

Parameter	Category	Median TMB	<i>P</i> value (Wilcoxon test)
Age	<40 years	1.11	<0.001
	≥ 40 years	1.74	
Stage	0–I	1.43	0.086
	IIA or higher	1.62	
Histol. grade	G2	1.56	0.826
	G3	1.52	
Treatment arm	Placebo	1.59	0.672
	Durvalumab	1.47	
Window treatment	No	1.70	0.303
	Yes	1.46	
TILs	<60%	1.61	0.190
	$\geq 60\%$	1.35	
PD-L1	Negative	1.43	0.989
	Positive	1.59	
Response	RD	1.39	0.005
	pCR (ypT0 ypN0)	1.87	

pCR, pathological complete remission; PD-L1, programmed death-ligand 1; RD, residual disease; TILs, tumor infiltrating lymphocytes.

Significant *P* values are given in bold.

higher TMB (median with pCR 1.87 versus 1.39 without pCR; $P = 0.005$). [Figure 1](#) shows the distribution of TMB values of patients with and without a pCR. Interestingly, a high TMB among pCR cases was especially seen in the placebo (chemotherapy alone) treatment arm ([Figure 1](#)).

As shown in [Table 2](#), we detected a predictive value of continuous TMB for pCR among all patients both in univariate (OR 1.62, 1.20–2.20; $P = 0.002$) and multivariate logistic regression (including age, stage, grading, stromal TILs, PD-L1 status, and window treatment, OR 2.06, 1.33–3.20; $P = 0.001$). When we separately analyzed the two treatment arms, we detected a significant positive correlation between TMB and pCR in both arms in multivariate analyses; in the univariate analysis we found a strong trend in the durvalumab arm and a significant correlation in the placebo arm ([Table 2](#)). We did not observe an interaction with the treatment arm ($P = 0.439$ and $P = 0.436$ for univariate and multivariate analysis, respectively, [Table 2](#)).

We also present results for TMB dichotomized using the upper tertile of our cohort (2.05 mut/Mb), which was similar to a previously published cutoff from a pan-cancer analysis of ICB response prediction.¹⁰ As shown in [Table 2](#), dichotomized TMB significantly predicted pCR among all patients (univariate OR 2.22, 1.11–4.43, $P = 0.024$; multivariate OR 3.45, 1.41–8.48, $P = 0.007$). Also in this analysis we found no significant interaction with the treatment arm ([Table 2](#)). After dichotomization of TMB at the top tertile, 50 patients had high TMB and 29 of these (58%) achieved a pCR, while 99 had low TMB and only 38 of these (38%) had a pCR ($P = 0.007$) ([supplementary Figure S2](#), available at *Annals of Oncology* online). In the durvalumab treatment arm, pCR rates were 17/27 (63%) and 19/47 (40%) for high and low TMB, respectively ($P = 0.028$), and in the placebo arm 12/23 (52%) for high, and 19/52 (37%) for low TMB ($P = 0.232$, [supplementary Figure S2](#), available at *Annals of Oncology* online).

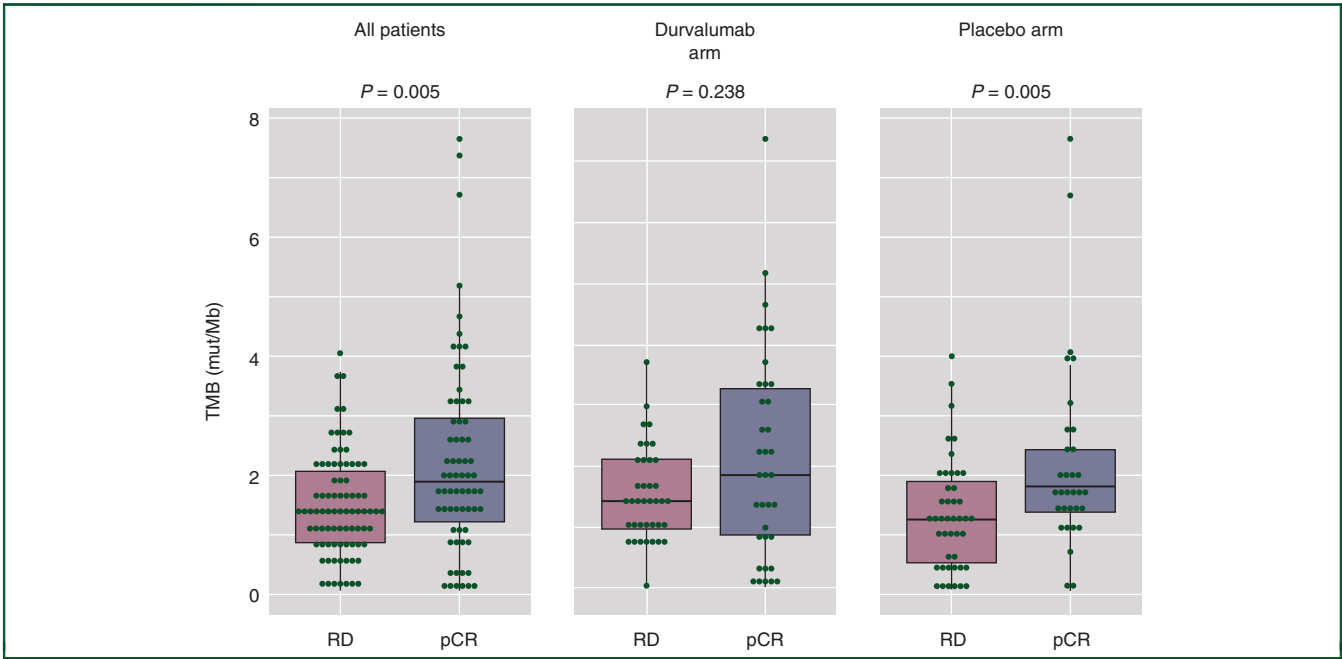


Figure 1. Association of pathological complete response and tumor mutational burden (TMB) in GeparNuevo. Distribution of TMB values in pretreatment samples of GeparNuevo patients stratified as either residual disease or pathological complete remission (pCR) among all patients and separately in the durvalumab and placebo arm, respectively.

Joint relationship of TMB and immune gene expression profile with pCR in GeparNuevo

We previously reported that TILs and immune GEP predicted pCR in the GeparNuevo study.^{3,25} For 136 patients, both WES and RNA-Seq data were available (supplementary Figure S1, available at *Annals of Oncology* online). Both a predefined immune GEP (GeparSixto immune signature) and TILs predicted pCR in this cohort in multivariate analysis (OR 1.73, 2.16–2.59, $P = 0.008$ and OR 1.32, 1.08–1.60, $P = 0.006$ for GEP and stromal TILs, respectively). Therefore, we also analyzed whether GEP or TILs correlate with continuous TMB. We found no significant correlation of either the immune GEP (Spearman’s rho +0.03, –0.14 to +0.19, $P = 0.771$) or TILs (Spearman’s rho –0.09, CI –0.25 to +0.07, $P = 0.269$) with TMB. In contrast, we could clearly detect that age, for example, was strongly correlated with

continuous TMB (Spearman’s rho +0.36, 0.21–0.49, $P < 0.001$). The observed independence of TMB and immune GEP or TILs from each other suggests that both factors may add predictive value for response, as has been suggested for other cancer types.¹⁰ Figure 2 demonstrates this contribution of both factors for TMB and GEP (see supplementary Figure S3, available at *Annals of Oncology* online, for individual treatment arms). In a multivariate model, both continuous parameters were independent significant predictors of pCR (supplementary Table S2, available at *Annals of Oncology* online). As shown in Figure 2B, the pCR rate among patients with both high TMB and high GEP in pretreatment samples was 82% (60% to 95%) compared with only 28% (16% to 43%) in the groups with both low TMB and GEP. When we used TILs instead of immune GEP, we obtained similar results with a pCR rate of 83% (36% to

Table 2. Predictive value of tumor mutational burden (TMB) for pathological complete remission (pCR) in GeparNuevo						
			All patients	Durvalumab	Placebo	Test for interaction
Continuous TMB (mut/Mb)	Univariate	n	149	74	75	0.439
		OR (95% CI)	1.62 (1.20–2.20)	1.45 (0.99–2.14)	1.87 (1.13–3.08)	
		P value	0.002	0.060	0.014	
	Multivariate ^a	n	133	64	69	
		OR (95% CI)	2.06 (1.33–3.20)	1.77 (1.00–3.13)	2.82 (1.21–6.54)	
		P value	0.001	0.049	0.016	
Dichotomized TMB (upper tertile)	Univariate	n	149	74	75	0.694
		OR (95% CI)	2.22 (1.11–4.43)	2.51 (0.95–6.64)	1.89 (0.70–5.12)	
		P value	0.024	0.065	0.208	
	Multivariate ^a	n	133	64	69	
		OR (95% CI)	3.45 (1.41–8.45)	4.66 (1.18–18.48)	2.21 (0.60–8.12)	
		P value	0.007	0.028	0.232	

CI, confidence interval; OR, odds ratio.
Significant P values are given in bold.
^a Including age, stage, grading, stromal tumor infiltrating lymphocytes, PD-L1 status, and window treatment.

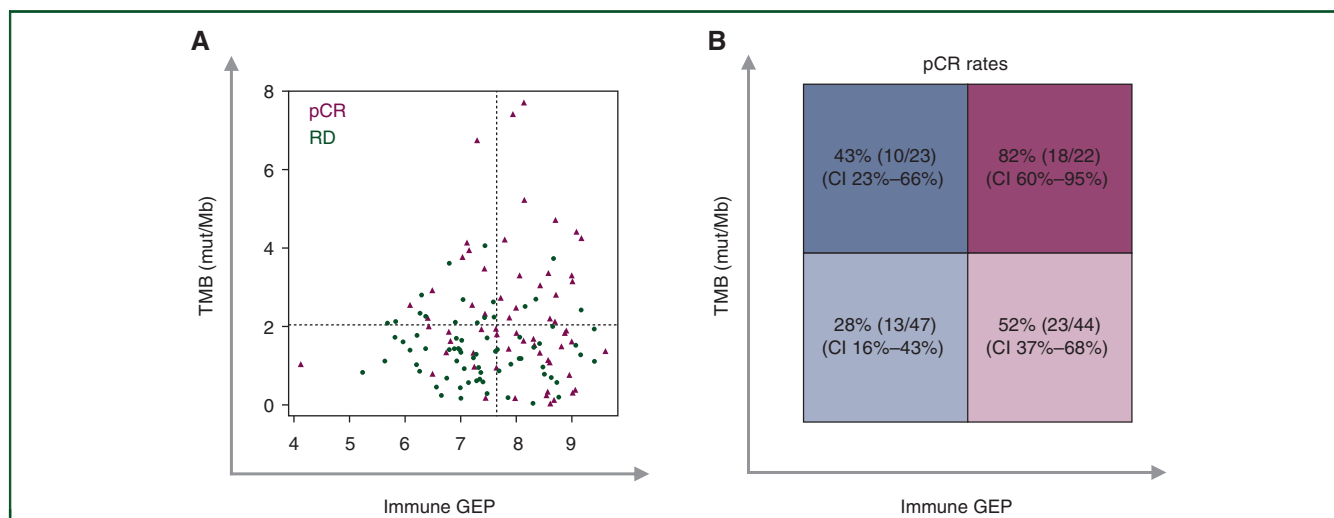


Figure 2. Joint relationship of tumor mutational burden (TMB) and immune gene expression profile (GEP) with pCR in GeparNuevo.

(A) Scatter plot of TMB and immune GEP in pretreatment biopsies of GeparNuevo patients colored by response [burgundy triangles, pathological complete remission (pCR); green circles, residual disease (RD)]. Cutoffs of median GEP and upper tertile of TMB are given by dashed vertical and horizontal lines, respectively. (B) pCR rates in percentages and 95% confidence intervals (CI) in subgroups defined by the cutoffs given as dashed lines in A.

100%) in the group with both high TMB and high TILs compared with 33% (23% to 44%) in the low/low group (supplementary Figure S4, available at *Annals of Oncology* online), and again both parameters were independent significant pCR predictors (supplementary Table S3, available at *Annals of Oncology* online).

TMB as a potential surrogate marker for biologically distinct groups of tumors

We next asked whether a high TMB may be a surrogate marker for a specific type of genomic alteration or for molecular subtypes, which could add mechanistic explanations for its association with response. We studied continuous scores of the mutational signatures 2, 3, 6, and 13 of Alexandrov et al.²⁰ and binary classifications of tumors characterized by specific driver mutations as well as RNA-based molecular subtypes (AIMS method).²³ We observed a strong positive correlation of TMB with continuous scores for mutational signature 2 (APOBEC-related, $\rho = 0.44$, 0.30–0.56, $P < 0.001$) and signature 3 [homologous recombination deficiency (HRD)-related, $\rho = 0.37$, 0.22–0.50, $P < 0.001$] and a weak negative correlation with signature 6 (MMR-related, $\rho = -0.18$, -0.33 to -0.02, $P = 0.030$). As shown in supplementary Table S4 (available at *Annals of Oncology* online), we also compared the median TMB in groups of tumors characterized by specific driver mutations or molecular subtype. We found no difference in median TMB between TNBC of the ‘basal-like’ or ‘HER2-enriched’ subtype. In contrast, we detected significant higher TMB values in tumors with mutations in *BRCA2*, *TP53*, and *ARID1A* ($P = 0.004$, $P = 0.011$, and $P = 0.040$, respectively) and a trend for *BRCA1*, *NOTCH1*, and *PTEN* ($P = 0.091$, $P = 0.063$, and $P = 0.099$, respectively) but no differences for *ATM*, *CCNE1*, *MYC*, and *PIK3CA* (supplementary Table S4, available at *Annals of Oncology* online). Moreover, a predefined panel of 16 genes involved

in HRD identified 45 mutated tumors with significantly higher TMB (1.89 versus 1.37 mut/Mb, $P < 0.001$; supplementary Table S4, available at *Annals of Oncology* online).

A recent paper identified specific copy number gains in immune-related genes in those breast cancers with high TMB but poor immune cell infiltration.¹⁵ When we analyzed GeparNuevo samples characterized by these copy number gains, we observed a non-significantly higher TMB (supplementary Figure S5A, available at *Annals of Oncology* online). However, we were not able to detect an association with a reduced immune infiltration in high TMB patients in our sample cohort (supplementary Figure S5B, available at *Annals of Oncology* online).

DISCUSSION

The predictive value of TMB for response to ICB has been reported for several cancers, mostly in the metastatic setting.^{6–9} However, its value for response to neoadjuvant therapy in early TNBC is not known, especially in combination with other parameters as immune GEP or TILs either for ICB or for neoadjuvant chemotherapy alone. We analyzed the predictive value of TMB for pCR (ypT0 ypN0) both alone and in combination with an immune GEP or TILs in a randomized neoadjuvant ICB trial in TNBC. We found that both TMB and immune GEP or TILs add independent value for pCR prediction. Interestingly, this result was obtained for both arms of the trial, that is for ICB in combination with chemotherapy and for chemotherapy alone.

These are novel results for breast cancer. But knowledge about TMB and response to chemotherapy is also limited in other cancers. In ovarian cancer TCGA data, a positive prognostic effect of high TMB was confined to *BRCA1/2* mutated cases.²⁶ Among more than 5000 different metastatic cancers without ICB treatment, no prognostic effect could be detected.⁹ In contrast, in colon cancer treated with

chemotherapy, high TMB was associated with improved survival, independent of its association with hypermutated tumors.²⁷ In the latter case, high TMB could be a surrogate for increased immune cell infiltration. However, our study indicates that both parameters seem to contribute independent prognostic value.

TMB values among more than 100 000 tumors differed largely (0–1241 mut/Mb) with medians between different cancer types from 0.8 up to 45.2 mut/Mb.^{20,28} Moreover, different platforms and analysis pipelines influence numerical TMB values.^{9,28} Therefore development of TMB cutoffs is challenging.^{9,10,28} While targeted sequencing approaches used different fixed cutoff values, most WES studies applied percentiles of cohorts.^{6–10,28} We have applied both continuous TMB and a dichotomized score. The upper tertile we used includes samples with ≥ 95 total mutations, not far from the optimized cutoff of >100 mutations in a pan-cancer analysis of ICB.¹⁰ In fact, 41 TNBC from our study (28%) surpass this published cutoff.¹⁰ In a recent study of breast cancers from TCGA, the prognostic value of an immune GEP was detected only in samples with TMB >1.63 mut/Mb.¹⁵ Again, this cutoff value is similar to the upper tertile of TNBC (1.75 mut/Mb) in the respective TCGA dataset. Despite clinical demand for dichotomized TMB cutoffs, optimization resembles that of other continuous parameters such as TILs and Ki67 in breast cancer^{19,29} and the quest for the ‘right cutoff’ may be difficult.

Importantly, our results suggest that a combination of TMB and immune GEP can increase the precision of response prediction. We also obtained similar results when we replaced GEP with TIL scoring, which could be easier to incorporate into routine settings. TMB did not correlate significantly with TILs or with immune GEP and contributes independent prognostic information.

We were also interested in whether TMB represents a surrogate for distinct biological groups of tumors. We observed no difference between TNBC of ‘basal-like’ or ‘HER2-enriched’ subtype but found three driver genes significantly associated with TMB: *BRCA2*, *TP53*, and *ARID1A* (supplementary Table S4, available at *Annals of Oncology* online). Interestingly, these genes were also among the top 7% of TMB-associated genes in a previous comprehensive study of more than 100 000 tumors.²⁸ However, we detected no predictive value for pCR of these individual genes: *BRCA2* $P = 0.977$, *TP53* $P = 0.408$, *ARID1A* $P =$ not available, HRD panel $P = 0.407$. This suggests that TMB captures predictive information beyond individual driver genes.

The strengths of our study include the use of samples from a prospectively randomized controlled trial with central TNBC confirmation and TIL assessment. In addition, we focused on a biologically motivated, limited set of pre-specified variables and our WES dataset represents the largest so far for early breast cancers with neoadjuvant ICB treatment. However, our study clearly has limitations. First, the sample size is still small and the complete set of molecular data was only available for a subset of the trial patients, which could introduce bias even if we detected no

differences in baseline parameters. Second, the clinical impact of the study is limited by the fact that neither TMB or immune GEP or TILs had a specific predictive effect for the addition of immunotherapy but identified patients with a high chance of pCR in both arms of the trial. Finally, a proper validation dataset for our findings is missing.

In conclusion, our study shows for the first time that TMB and immune infiltration add independent value for pCR prediction both for ICB and for chemotherapy. Our results merit validation and recommend further study of both parameters to individually tailor therapies in breast cancer.

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DISCLOSURE

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Supplementary Material

Supplementary Methods

Supplementary information is available online at <https://github.com/tkarn/G9-TMB>.

Whole Exome sequencing

Whole exome sequencing was conducted on fresh-frozen pre-therapeutic core biopsies and patient-matched blood samples. Tumor purity and cellularity can have impact on the detection of mutations from whole exome sequencing and thereby potentially confound TMB estimation (see below). Therefore, selection of tumor tissue was performed by experienced pathologists to obtain samples with high cellularity (>50%). For each sample 10x 5µm cryosections were used for whole exome sequencing with *Illumina HiSeq 4000* (Illumina Inc., San Diego, CA, USA) for 2 x 100 paired end reads. High quality data was obtained for 149 (85.6%) of the 174 patients from the trial (Supplementary Figure S1). The sequence reads were mapped to human reference genome hg19 using *BWA* (Burrows-Wheeler aligner) [1]. *Samtools* and *Bedtools* were used to calculate mapping rate and coverage for quality control (QC) [2, 3]. *Mutect2* [4] was used to make somatic variant calls, including single-nucleotide-variant (SNV) and small-insertion-deletion (indel). Gene and functional annotation of the called SNVs and indels was carried out using *ANNOVAR* [5]. We used COSMIC v86 and dbSNP v152 variants as white and black lists for *MuTect2* respectively. This resulted in 34,082 mutations in 10,969 genes. 96 genes from a black list and 16,471 SNVs flagged in germline by *ANNOVAR* were excluded. Of the remaining 15,682 mutations 3,368 synonymous SNV were filtered resulting in a final list of 12,314 mutations. For somatic copy number alterations *pure CN* [6] was used.

Tumor mutational burden and mutational signatures

For TMB calculation we used the final list of 12,314 non-synonymous SNVs and indels together with an effective DNA coverage of 46 Mb to determine mutations per Mb for each sample. Mutational signatures were identified as described by Alexandrov et al [7]. R package *SomaticSignatures* [8] was employed to estimate the proportion of each sample's mutations that have been assigned to each of the 21 mutational signatures.

Based on our previous analyses of TNBC from TCGA we were concerned that high immune infiltration cells could bias towards a lower TMB, because of reduced tumor cell content. In that previous study we found a negative correlation between TMB and immune cell infiltration (despite $\geq 50\%$ tumor cell content mandatory for TCGA samples) [9]. However, in the GeparNuevo dataset we did not detect such negative correlation between immune cell gene expression and TMB (Fig. 2A). Moreover, in the TCGA dataset we had also found a positive correlation between TMB and tumor cellularity (as measured by ASCAT) [9]. Again, in the GeparNuevo data we did not observe such a correlation between TMB and purity (as measured by pureCN, Spearman's $\rho=0.006$). Thus, despite no systematic data on histological assessment of tumor content was available, these observations argue against a strong bias of TMB by tumor cellularity in the GeparNuevo dataset.

RNA sequencing

RNA sequencing was performed on formalin-fixed paraffin-embedded (FFPE) tissue using a *HTG EdgeSeq* instrument (HTG Molecular Inc, Tucson, AZ, USA) with the *HTG EdgeSeq Oncology Biomarker Panel* (2549 genes) according to the manufacturer's instructions. Tumor area was marked on an H&E stained slide and the area of invasive breast cancer recorded. From a corresponding unstained slide 15 mm² tissue was scraped and used for library preparation. The method is based on an RNA-extraction-free chemistry and a nuclease protection assay [10]. Libraries were quantified, pooled and sequenced on an *Ion Torrent S5* instrument (Thermo Fisher Scientific, Waltham, MA, USA). Count tables were generated using the *HTG parsing tool*. For quality control, we transformed the reads to counts-per-million and calculated the mean of five negative and four positive internal controls for each sample. We repeated processing for a sample if the mean of its positive controls was below two standard deviations (SDs) of the grand mean across all samples or if the mean of its negative controls were above two SDs from the grand mean (Supplementary Figure S6). The data was median normalized within a sample and across the experiment by calculating a scaling factor for each sample as the median gene expression value for each sample-gene count adjusted by the geometric mean over all genes. RNA sequencing data from pre-therapeutic cores were available for 159 of the 174 patients (Supplementary Figure S1).

Molecular subtyping by AIMS from RNA-Seq

Molecular subtyping from RNA-Seq was performed using the *Absolute Intrinsic Molecular Subtyping* (AIMS) method by applying the R package *AIMS*. Of the 136 patients with both WES and RNA-Seq data, 82 (60.3%) were classified as basal-like, 50 (36.7%) as HER2-enriched, 3 (2.2%) as normal-like, and 1 (0.7%) as luminal A. The *HTG-EdgeSeq Oncology Biomarker Panel* encompasses 2549 genes, which includes 41 of the 100 gene pairs proposed as rules for the AIMS algorithm. The respective list of available gene pairs is given in [Supplementary Table S5](#) with 6-10 rules per subtype. In the original AIMS publication [11] the agreement between AIMS and PAM50 classification dropped below 10 rules per subtype. Therefore, the lower number of rules available in our study may be one explanation for the relatively high frequency of HER2-enriched cases (36.7%) among TNBC in our cohort in contrast to PAM50 data from the literature e.g. 7.8% [12].

Immune gene expression profiles (GEP) from RNA-Seq

We evaluated a predefined immune gene expression profile (GEP) predictive for neoadjuvant response that was created from a list of genes we previously identified in the GeparSixto study (“GeparSixto immune signature”: CXCL9, CCL5, CD8A, CD80, CXCL13, IDO1, PDCD1, CD274, CTLA4, FOXP3) [13]. The genes CD21 and IGKC were omitted because they were not covered by the HTG-EdgeSeq Oncology Biomarker Panel. The immune GEP was calculated as mean of the expression of the genes from the signature. We also verified substantial agreement of this pre-specified signature with various other gene signatures for lymphocyte infiltration (Pearson correlation values 0.74-0.97, [Supplementary Figure S7](#)).

Aggregation of WES and HTG-RNA-Seq data

The analyses of the pseudonymized genomic datasets were performed fully blinded to any clinical or pathological sample information. The final blinded WES and HTG-RNA-Seq datasets were transferred to GBG-headquarters. WES and RNA-Seq were available for 149 and 159 patients, respectively, with both data available for 136 patients ([Supplementary Figure S1](#)). A comparison of the complete trial cohort, the WES cohort, and the WES+RNA-Seq cohort is provided in [Supplementary Table S1](#). An integrated view of all measures is presented in [Supplementary Figure S8](#).

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Supplementary Tables

Suppl. Table S1: Clinical data of the complete study cohort vs. the WES cohort and the WES+RNA-Seq cohort.

Parameter	Category	Complete study cohort	WES cohort	P-value	WES + RNA-Seq cohort	P-value
Number of patients		174	149		136	
Age	< 40yr	47 (27.0%)	37 (24.8%)	0.144	33 (24.3%)	0.149
	≥ 40yr	127 (73.0%)	112 (75.2%)		103 (75.7%)	
Stage	0-I	61 (35.1%)	51 (34.2%)	0.652	47 (34.6%)	0.848
	IIA or higher	113 (64.9%)	98 (65.8%)		89 (65.4%)	
Histol. Grade	G2	29 (16.7%)	22 (14.8%)	0.142	21 (15.4%)	0.461
	G3	145 (83.3%)	127 (85.2%)		115 (84.6%)	
TILs	< 60%	149 (85.6%)	128 (85.9%)	0.762	116 (85.3%)	1.000
	≥ 60%	25 (14.4%)	21 (14.1%)		20 (14.7%)	
PD-L1	negative	20 (12.7%)	16 (12.0%)	0.526	14 (11.3%)	0.382
	positive	138 (87.3%)	117 (88.0%)		110 (88.7%)	
Treatment arm	Placebo	86 (49.4%)	75 (50.3%)	0.667	67 (49.3%)	1.000
	Durvalumab	88 (50.6%)	74 (49.7%)		69 (50.7%)	
Window treatment	No	57 (32.8%)	48 (32.2%)	0.818	47 (34.6%)	0.435
	Yes	117 (67.2%)	101 (67.8%)		89 (65.4%)	
Response	RD	89 (51.1%)	82 (55.0%)	0.017	72 (52.9%)	0.463
	pCR (ypT0ypN0)	85 (48.9%)	67 (45.0%)		64 (47.1%)	

Suppl. Table S2: Logistic regression of pCR using TMB and immune GEP

<i>Bivariate</i>	N	136
TMB (mut/Mb)	OR (95% CI)	1.77 (1.24-2.51)
	P-value	0.001
Immune GEP	OR (95% CI)	1.81 (1.22-2.68)
	P-value	0.003
<i>Multivariate</i> [#]	N	124
TMB (mut/Mb)	OR (95% CI)	2.06 (1.34-3.16)
	P-value	0.001
Immune GEP	OR (95% CI)	1.58 (1.03-2.41)
	P-value	0.036

[#] including age, stage, grading, and window treatment (PD-L1 status was not included as additional covariate since PD-L1 is included in GEP)

Suppl. Table S3: Logistic regression of pCR using TMB and TILs

<i>Bivariate</i>	N	149
TMB (mut/Mb)	OR (95% CI)	1.78 (1.27-2.50)
	P-value	<0.001
TILs (per 10 %)	OR (95% CI)	1.38 (1.27-2.50)
	P-value	<0.001
<i>Multivariate</i> [#]	N	133
TMB (mut/Mb)	OR (95% CI)	2.06 (1.33-3.20)
	P-value	0.001
TILs (per 10 %)	OR (95% CI)	1.32 (1.08-1.61)
	P-value	0.006

[#] including age, stage, grading, PD-L1 status, and window treatment

Suppl. Table S4: Correlation of TMB with molecular subtypes and genomic alterations

Parameter	Category (n)	Median TMB	P-value [§] (Wilcoxon test)
Molecular subtype (AIMS) [†]	Basal-like (82)	1.49	0.299 0.512
	HER2-enriched (50)	1.60	
	other (4)	1.25	
ATM	mut (6) wt (143)	1.44 1.52	0.931
ARID1A	mut (5) wt (144)	2.46 1.47	0.040
BRCA1	mut (20) wt (129)	1.85 1.43	0.091
BRCA2	mut (9) wt (140)	2.70 1.43	0.004
CCNE1	amp (11) wt (138)	1.30 1.53	0.569
NOTCH1	mut (5) wt (144)	2.50 1.50	0.063
MYC	amp (39) wt (110)	1.43 1.56	0.269
PIK3CA	mut/amp (17) wt (132)	1.39 1.56	0.900
PTEN	mut/del (17) wt (132)	1.87 1.47	0.099
TP53	mut (103) wt (46)	1.65 1.14	0.011
HRD-panel [#]	mut/del (45) wt (104)	1.89 1.37	<0.001

[§] P-values not corrected for multiple testing

[†] Molecular subtype from RNA-seq was available only for 136 of the 149 samples with WES (see Supplementary Figure S1).

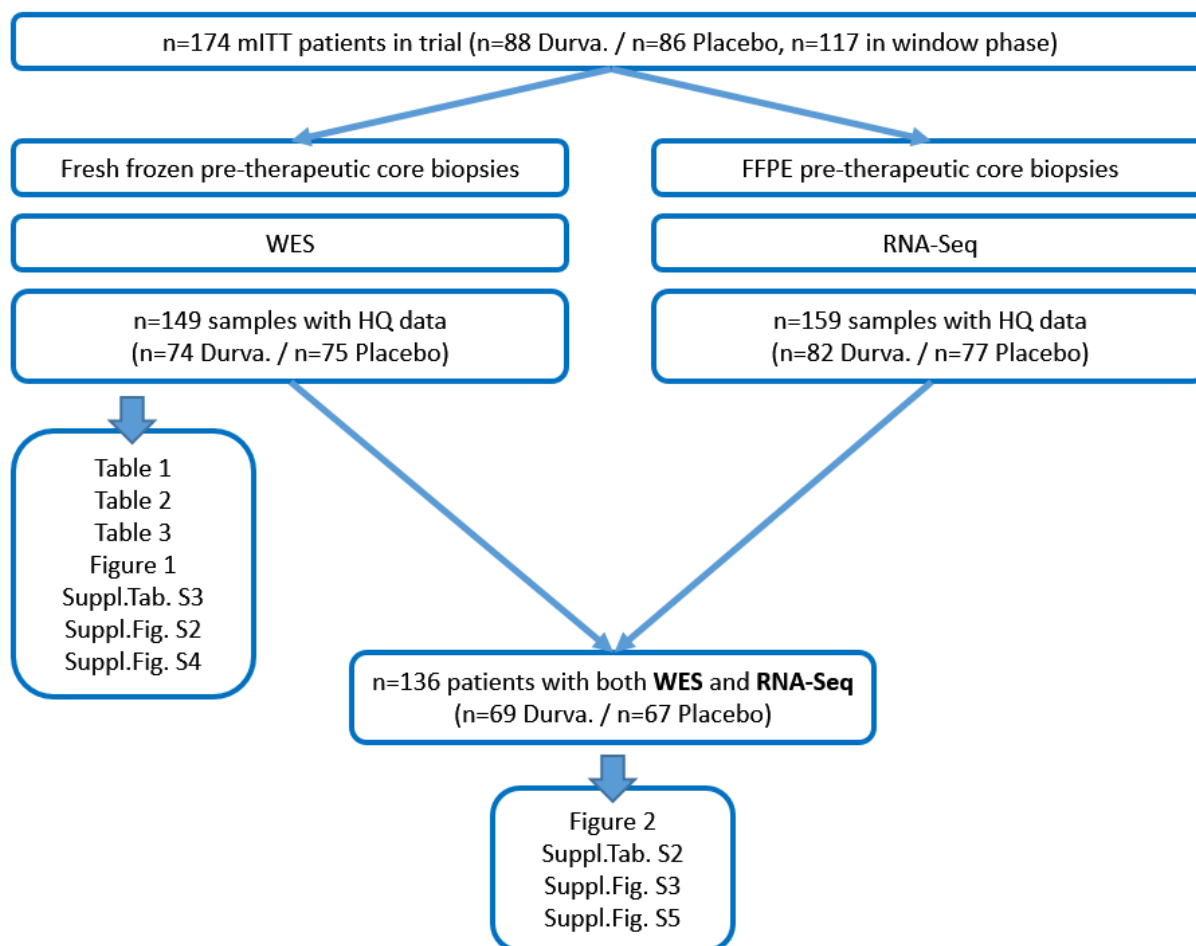
[#] The predefined HRD-panel encompassed: BRCA1, BRCA2, ATM, PALB2, BARD1, DRIP1, RAD51B, RAD51C, RAD51D, FAAP20, CHECK2, FAN1, FANCE, FANCM, POLQ, NBN

Suppl. Table S5: AIMS gene pairs included in *HTG EdgeSeq Oncology Biomarker Panel*

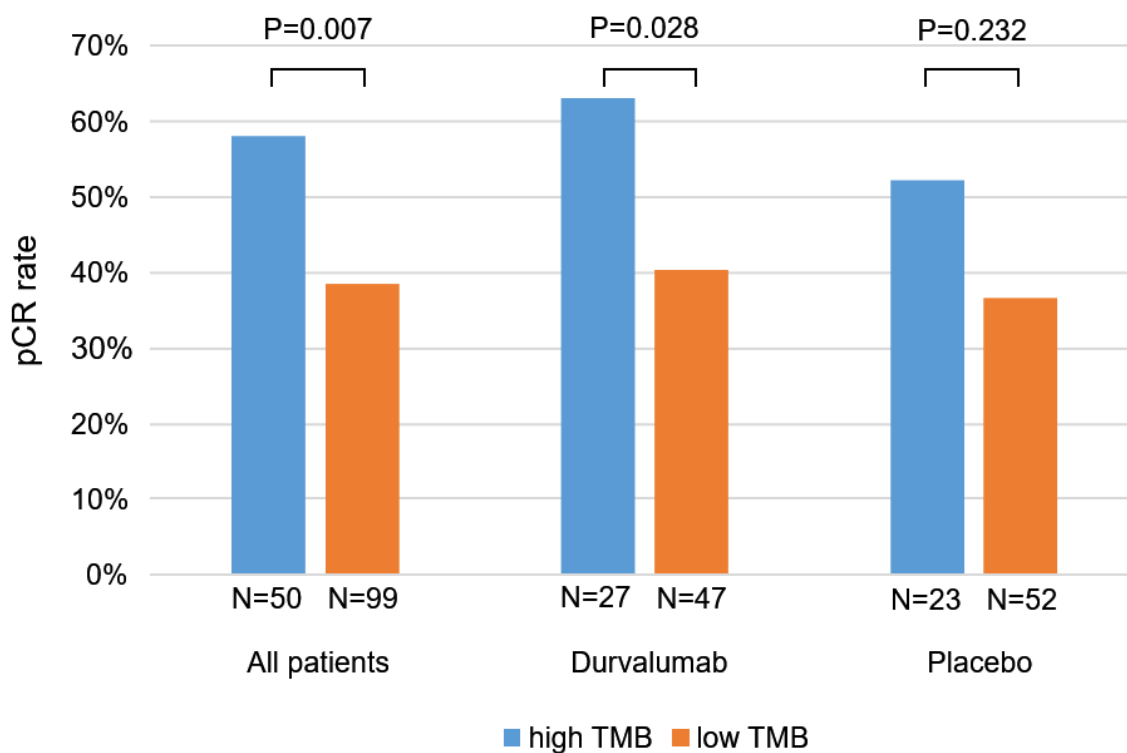
Subtype	rank [†]	gene	Entrez	gene	Entrez
BasalL	1	FOXA1	3169	< CDC20	991
BasalL	2	TFF3	7033	< KIF2C	11004
BasalL	3	CA12	771	< PTTG1	9232
BasalL	4	ESR1	2099	< FOXC1	2296
BasalL	6	NAT1	9	< ASPM	259266
BasalL	9	DNAJC12	56521	< MELK	9833
BasalL	17	KRT18	3875	< CSTB	1476
BasalL	19	SPDEF	25803	< TNFRSF21	27242
HER2E	2	SERPINA3	12	< ERBB2	2064
HER2E	3	RBBP8	5932	< RRM2	6241
HER2E	6	STC2	8614	< FOXA1	3169
HER2E	13	SLC39A6	25800	< TSPAN13	27075
HER2E	14	ESR1	2099	< NQO1	1728
HER2E	15	SCUBE2	57758	< UBE2C	11065
HER2E	16	KIT	3815	< FGFR4	2264
HER2E	17	IRS1	3667	< MELK	9833
HER2E	18	KRT17	3872	< S100A8	6279
HER2E	20	FOXC1	2296	< CDC20	991
LumA	1	CCNB2	9133	< MAP2K4	6416
LumA	2	BIRC5	332	< HPN	3249
LumA	4	KIF2C	11004	< SHC2	25759
LumA	8	PRC1	9055	< GAMT	2593
LumA	9	MELK	9833	< APH1B	83464
LumA	12	TTK	7272	< FMO5	2330
LumA	17	CDKN3	1033	< MNAT1	4331
LumA	20	CKS2	1164	< STC2	8614
LumB	5	CRYAB	1410	< RACGAP1	29127
LumB	6	CDH3	1001	< PRC1	9055
LumB	11	MMP7	4316	< TOP2A	7153
LumB	14	CX3CL1	6376	< KIF2C	11004
LumB	16	GSN	2934	< ESR1	2099
LumB	20	ANXA3	306	< CEP55	55165
NormL	4	CEP55	55165	< COL17A1	1308
NormL	5	PCNA	5111	< CAV1	857
NormL	7	CKS2	1164	< ITM2A	9452
NormL	8	RRM2	6241	< IGF1	3479
NormL	10	CDKN3	1033	< TSPAN7	7102
NormL	11	TYMS	7298	< SFRP1	6422
NormL	12	MCM2	4171	< CDKN1C	1028
NormL	17	UBE2C	11065	< PTN	5764
NormL	20	RFC4	5984	< IGFBP6	3489

[†] Rank of the respective rule among the 20 rules for each subtype used by AIMS [11]

Supplementary Figures

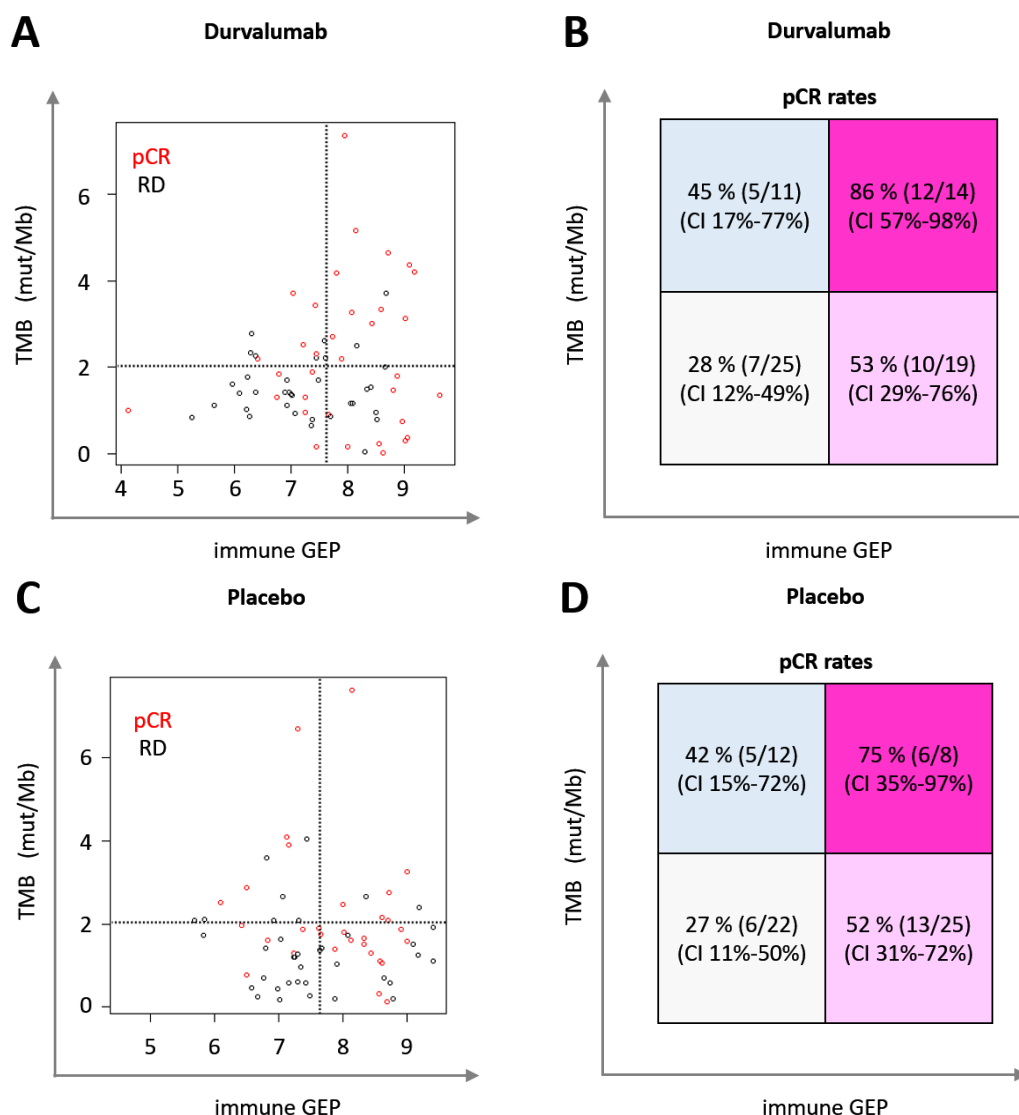


Supplementary Figure S1: CONSORT diagram of the flow of samples through the study



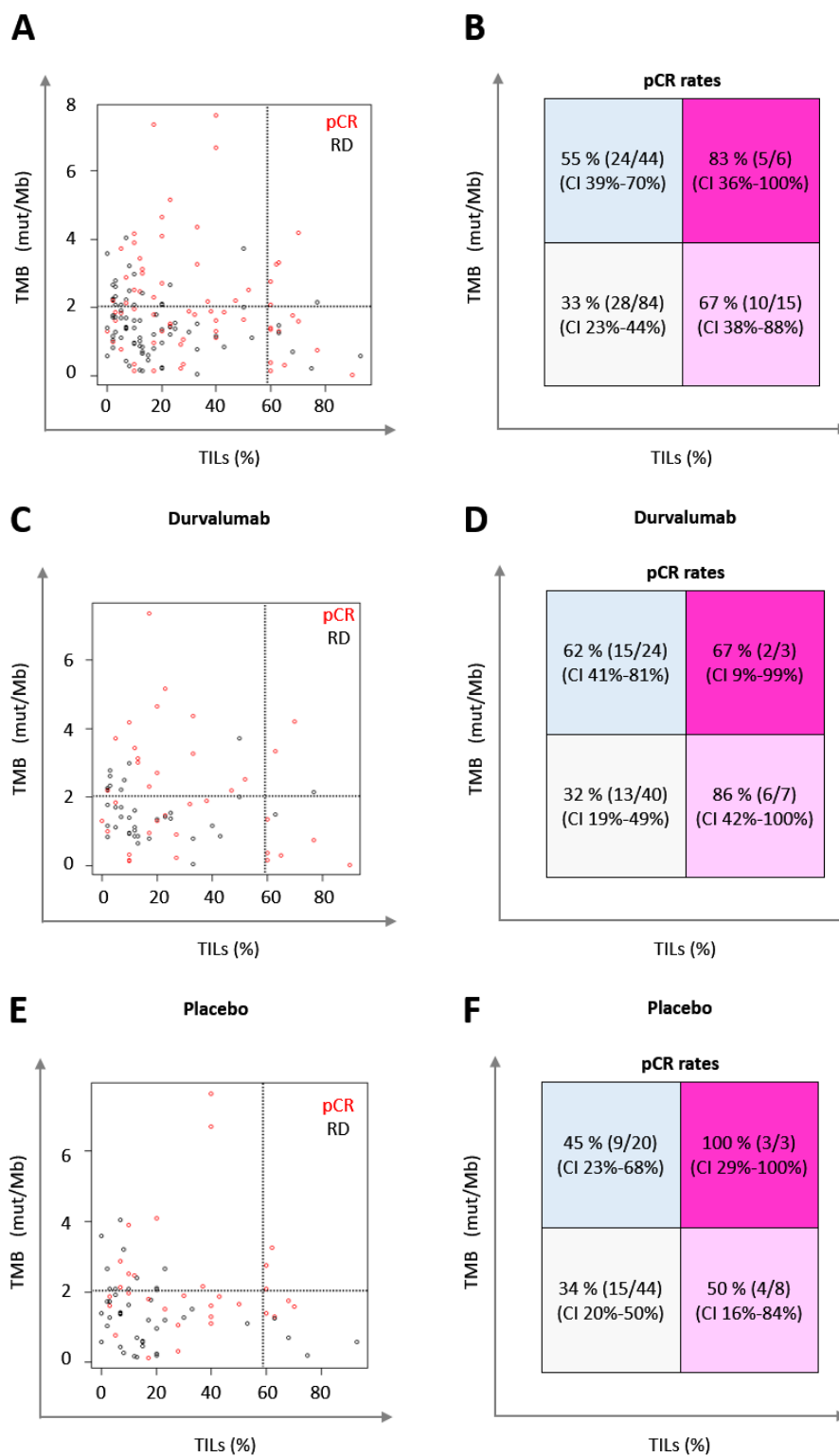
Supplementary Figure S2: Association of pathological complete response and TMB in GeparNuevo

Response rates (pCR) among patients with high (upper tertile) or low TMB in all GeparNuevo patients, and separately presented according to treatment arm (P-values from multivariate analysis).



Supplementary Figure S3: Joint relationship of TMB and immune gene expression profile with pCR separately within the treatment arms of GeparNuevo

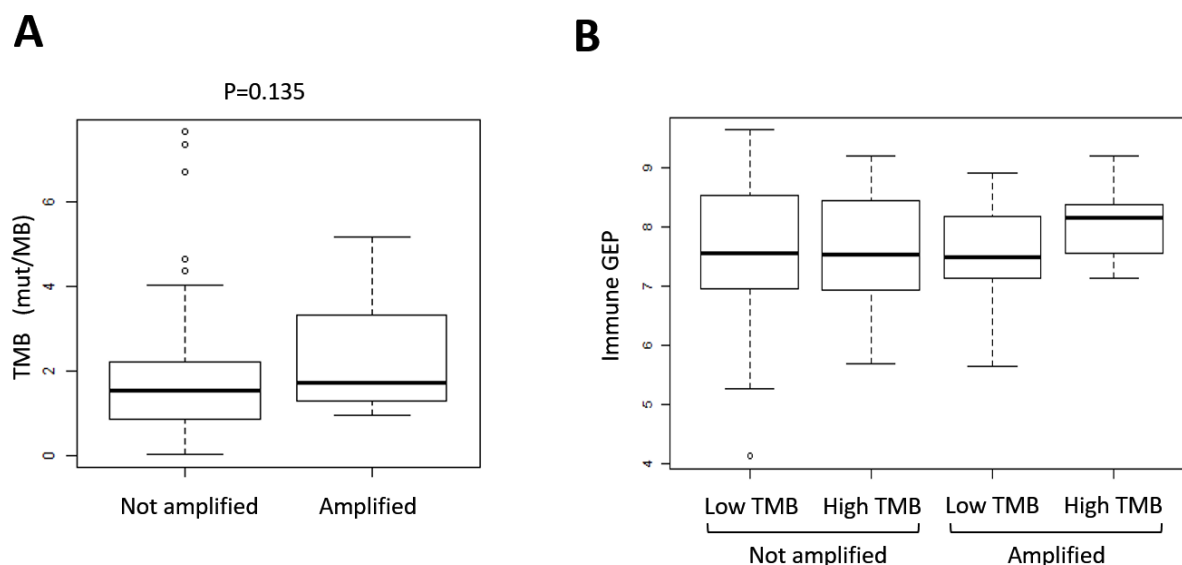
- Scatter plot of TMB and immune GEP in pretreatment biopsies of patients from the GeparNuevo durvalumab arm coloured by response. Median of GEP (of all samples from both arms) and upper tertile of TMB are given by dashed vertical and horizontal lines, respectively.
- pCR rates in percentages and 95% confidence intervals (CI) in subgroups defined by the cutoffs given as dashed lines in (A).
- Scatter plot of TMB and immune GEP in pretreatment biopsies of patients from the GeparNuevo placebo arm coloured by response. Median of GEP (of all samples from both arms) and upper tertile of TMB are given by dashed vertical and horizontal lines, respectively.
- pCR rates in percentages and 95% confidence intervals (CI) in subgroups defined by the cutoffs given as dashed lines in (C).



Supplementary Figure S4: Joint relationship of TMB and TILs with pCR in GeparNuevo

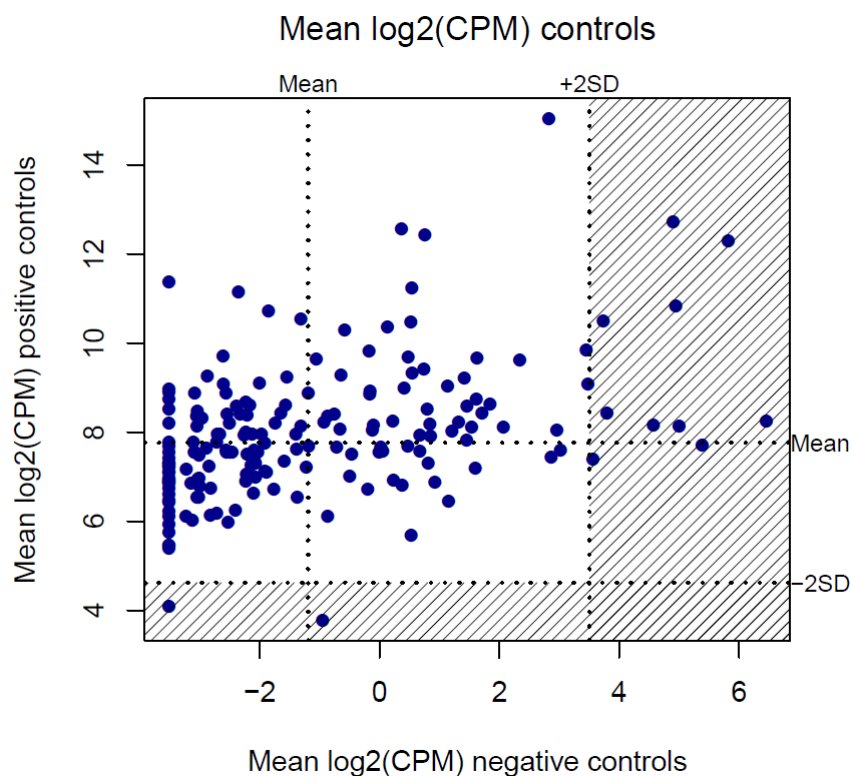
- A) Scatter plot of TMB and TILs in pretreatment biopsies of GeparNuevo patients coloured by response. The cutoff for “lymphocyte predominant breast cancer” (LBPC, $\geq 60\%$ TILs) and the upper tertile of TMB are given by dashed vertical and horizontal lines, respectively.
- B) pCR rates in percentages and 95% confidence intervals (CI) in subgroups defined by the cutoffs given as dashed lines in (A).

- C) Scatter plot of TMB and TILs in pretreatment biopsies of patients from the GeparNuevo durvalumab arm coloured by response. The LPBC cutoff ($\geq 60\%$ TILs) and the upper tertile of TMB are given by dashed vertical and horizontal lines, respectively.
- D) pCR rates in percentages and 95% confidence intervals (CI) in subgroups defined by the cutoffs given as dashed lines in (C).
- E) Scatter plot of TMB and TILs in pretreatment biopsies of patients from the GeparNuevo placebo arm coloured by response. The LPBC cutoff ($\geq 60\%$ TILs) and the upper tertile of TMB are given by dashed vertical and horizontal lines, respectively.
- F) pCR rates in percentages and 95% confidence intervals (CI) in subgroups defined by the cutoffs given as dashed lines in (E).



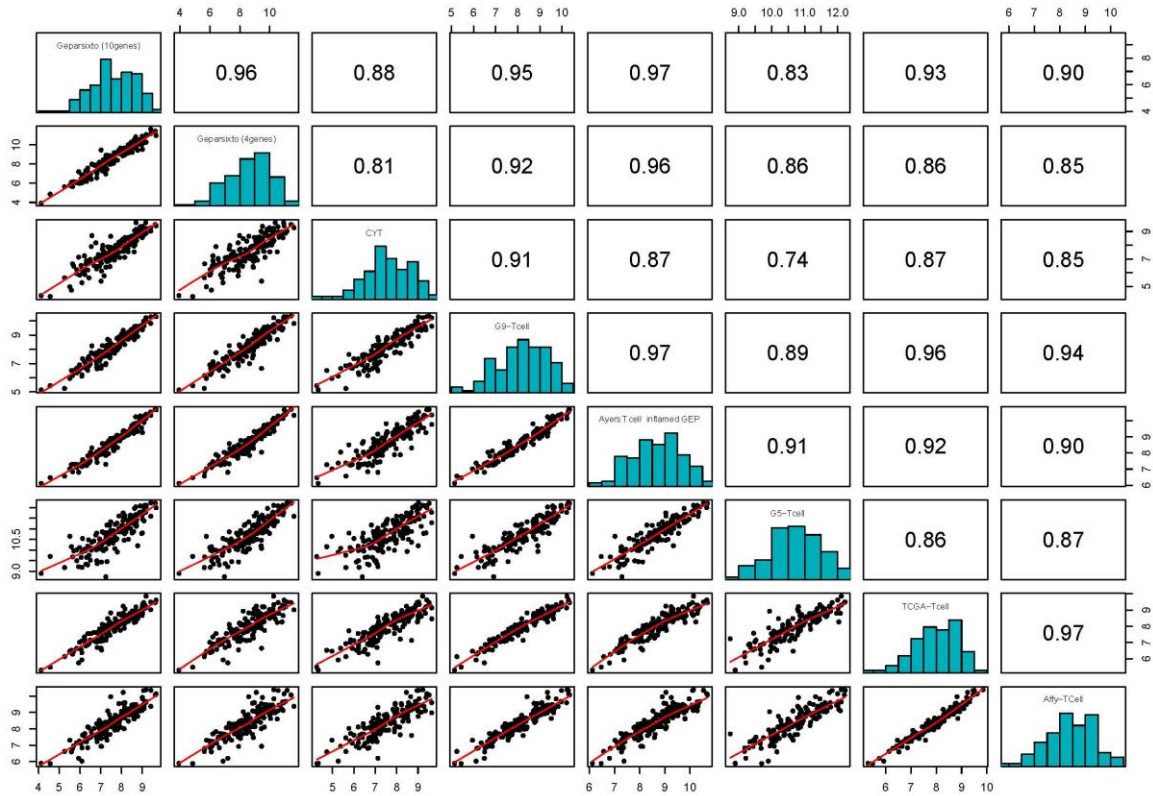
Supplementary Figure S5: Relationship of TMB and immune GEP in samples with specific copy number gains in GeparNuevo patients

- A) Box plot of TMB in sample groups characterized by specific copy number gains on Chr 1 (1q24.2, 1q25.1, 1q32.1). These amplifications have been previously associated with tumors that display low immune infiltration despite a high TMB (*Thomas et al. 2018 Oncoimmunology, PMID 30386679*). In the amplified cases a numerically but not significantly higher TMB was detected ($P=0.135$, Wilcoxon rank sum test).
- B) Box plot of immune gene expression in tumor groups classified both by the specific copy number gains (1q24.2, 1q25.1, 1q32.1) and TMB. No inverse association of TMB and immune GEP could be detected in the group characterized by the copy number gains.



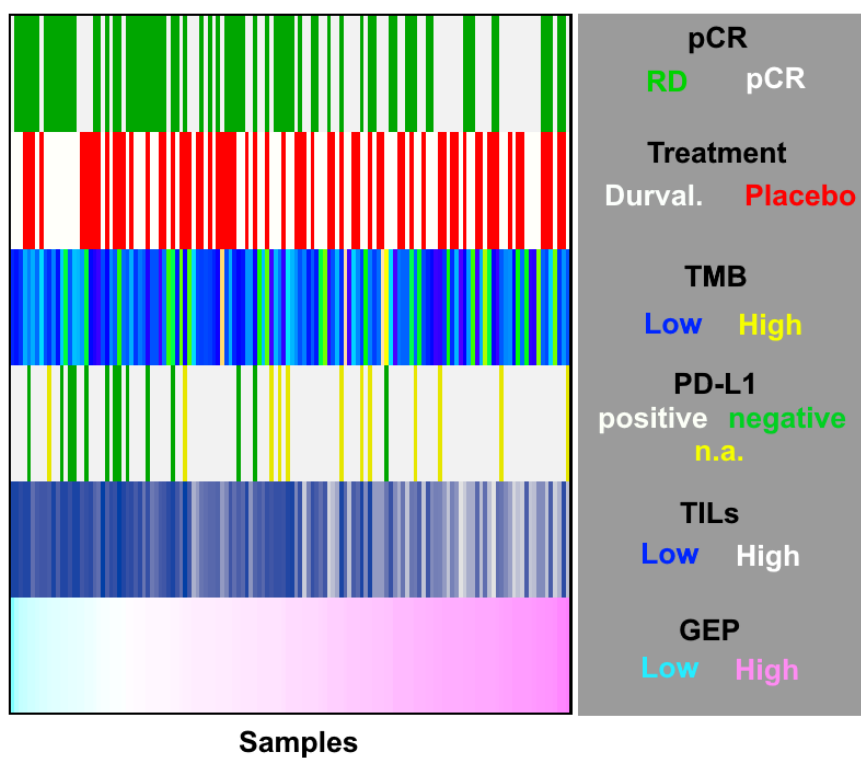
Supplementary Figure S6: Quality control of RNA sequencing data from HTG EdgeSeq

RNA-Seq reads were transformed to counts-per-million and the mean of five negative and four positive internal controls calculated for each sample. A sample was excluded if the mean of its positive controls was below two standard deviations (SDs) of the grand mean across all samples or if the mean of its negative controls were above two SDs from the grand mean (shaded regions in the figure).



Supplementary Figure S7: Correlation of different T-cell signatures from HTG EdgeSeq data

Several published gene signatures describing lymphocyte infiltration in breast cancer were obtained from HTG EdgeSeq data of the GeparNuevo samples. Results are compared by scatter plots and Pearson correlations presented. The following signatures were used: (1) The pre-specified 10 gene “GeparSixto immune signature” applied throughout all current analyses [13], (2) a 4 gene core of the GeparSixto immune signature, (3) the 2 gene “CYT” metric [14], (4) an 18 gene T-cell cluster from unsupervised clustering of the GeparNuevo HTG EdgeSeq data, (5) the “T-cell inflamed GEP” from Ayers et al. [15], as well as those genes available from the *HTG EdgeSeq Oncology Biomarker Panel* of the T cell clusters from studies of (6) FFPE-RNA-Seq of GeparQuinto [16], (7) RNA-Seq from TNBC of TCGA [9], and (8) Affymetrix microarrays of TNBC [17].



Supplementary Figure S8: Integrated view of parameters in GeparNuevo samples

Combined heatmap presentation to visualize the correlation of GEP, TILs, PD-L1 status, TMB, treatment arms, and pCR, among the 136 samples with data on both GEP and TMB. Samples have been ordered from left to right according to GEP as presented in the most lower panel. Colour codes for the different variables are given on the right.